

1.6 Bases

We have reached what many consider the “Goldilocks Zone” of Linear Algebra.

If a set is too small, it won’t span the whole space—you’ll be left with points you can’t reach. If a set is too large, it will be linearly dependent—you’ll have redundant vectors cluttering your workspace. A **basis** is a set that is “just right.” It is large enough to build everything, but small enough that every piece is essential.

Because a basis has no redundancy, it gives us something incredible: a **unique address system**. In an abstract vector space, it’s hard to tell someone where a vector is. But once we pick a basis, every vector can be described by a unique list of numbers—its coordinates. A basis is the bridge that allows us to turn abstract geometry into concrete arithmetic.

1.6.1 Definition and Standard Bases

Definition 1.6.1 — Basis

A subset B of a vector space V is called a **basis** for V if:

1. B is linearly independent;
2. B spans V .

Example. (Standard Bases)

Commonly used standard bases for familiar vector spaces are listed below:

- The **standard basis** for F^n is $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, where $\mathbf{e}_i$ has a 1 in the i -th position and 0 elsewhere.
- The standard basis for $F[x]_{\leq n}$ is $\{1, x, x^2, \dots, x^n\}$.
- The standard basis for $M_{m \times n}(F)$ is the set of matrices $\mathbf{E}_{ij}$ having a 1 at entry (i, j) and 0 elsewhere.

1.6.2 Unique Representation and Coordinates

The most important property of a basis is that every vector in the space has a unique “address” relative to it.

Theorem 1.6.1 — Unique Representation

Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Then every vector $\mathbf{v} \in V$ can be written as a linear combination of elements in B in **exactly one way**.

Proof. Since B spans V , there exist scalars a_i such that $\mathbf{v} = \sum a_i \mathbf{v}_i$. Suppose there is